Cloud Application Deployment

Design and Creative Technologies

Torrens University, Australia

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Subject Code: CCF 501

Subject Name: Cloud Computing Fundamentals

Assessment No.: 3

Title of Assessment: Cloud Application Deployment

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Date: April 2026

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Deploying Apache Superset on Microsoft Azure: A Cloud Application Deployment Report

1. Introduction

This report documents the deployment of Apache Superset – an open-source data exploration and visualization platform – on Microsoft Azure. Superset was selected for its direct relevance to data engineering workflows and its Python-native architecture, which aligns with the author's professional background and academic trajectory. Azure was chosen as the cloud provider to build familiarity ahead of the AZ-900 and DP-900 certification pathway, as recommended during academic advising in March 2026. The deployment covers the provisioning of a virtual machine, virtual network, firewall security policy, and a fully operational Superset instance accessible via public URL.

1.1. Background

In a traditional IT environment, standing up a data visualization platform would require procuring physical server hardware, configuring a local network, applying firewall rules at the rack level, and maintaining that infrastructure indefinitely. For an individual developer or small team, that overhead makes self-hosted analytics tooling impractical outside enterprise settings.

Cloud computing removes that barrier. The National Institute of Standards and Technology (NIST) defines cloud computing as on-demand network access to a shared pool of configurable computing resources, characterized by five essential properties: on-demand

self-service, broad network access, resource pooling, rapid elasticity, and measured service (Mell & Grance, 2011). These properties shift infrastructure from a capital expenditure to an operational one – organizations pay only for what they use, without fixed-capacity procurement or physical hardware management (McHaney, 2021).

This deployment uses an **Infrastructure as a Service (IaaS)** model: Azure provides the virtual machine, virtual network, and compute layer; the operating system, container runtime, and application stack are managed by the author. The deployment model is **public cloud** – hosted on Azure's shared global infrastructure and secured through network-level access policies. Where a traditional server setup requires weeks of procurement and physical installation, the same outcome here was achieved through a browser-based portal at effectively zero capital cost (IBM, n.d.-a).

2. Body

2.1. Service Provider Selection Rationale

Microsoft Azure was selected for three reasons. First, its data and analytics ecosystem – Azure SQL, Synapse Analytics, and Data Factory – directly complements Superset's role as a query and visualization layer, relevant to current data engineering work and the BDA601 subject ahead. Second, Azure is the target certification platform (AZ-900 → DP-900), making hands-on deployment a practical study activity. Third, Azure's free-tier eligibility lowers the barrier; the deployment uses B2s (2 vCPU, 4 GB RAM) as Superset's Docker stack requires more RAM than the free B1s allows, covered by student credits.

Criterion	AWS	Microsoft Azure	GCP
Free tier VM	EC2 t2.micro (750h/month)	B1s (750h/month)	e2 micro (always free)
Cost model	PAYG; Reserved (75% off); Spot	PAYG; Reserved (72% off); Spot	PAYG; Reserved (70% off); Spot
Resource elasticity	Auto Scaling Groups + ELB	VM Scale Sets + ALB	Managed Instance Groups + CLB
Data/ML integration	RDS, Redshift, SageMaker	Azure SQL, Synapse, Azure ML	BigQuery, Vertex AI
Cert alignment (this enrolment)	x	AZ-900 / DP-900	x

Table 1: Cloud provider comparison for Apache Superset deployment.

2.2. Deployment Model and Architecture Diagram

The deployment follows a **public cloud IaaS** model. All resources are provisioned within a single Azure Resource Group, logically isolated by a Virtual Network (VNet) with a dedicated application subnet. A Network Security Group (NSG) enforces inbound traffic rules, restricting access to SSH (port 22, source IP restricted) and the Superset web interface (port 8088). HTTP (port 80) and HTTPS (port 443) are intentionally excluded – Superset is accessed directly on port 8088, and TLS termination is identified as a future improvement in Section 2e. Apache Superset runs inside a Docker Compose stack on an Ubuntu 22.04 virtual

machine, with PostgreSQL as the metadata database and Redis as the caching and Celery task broker.

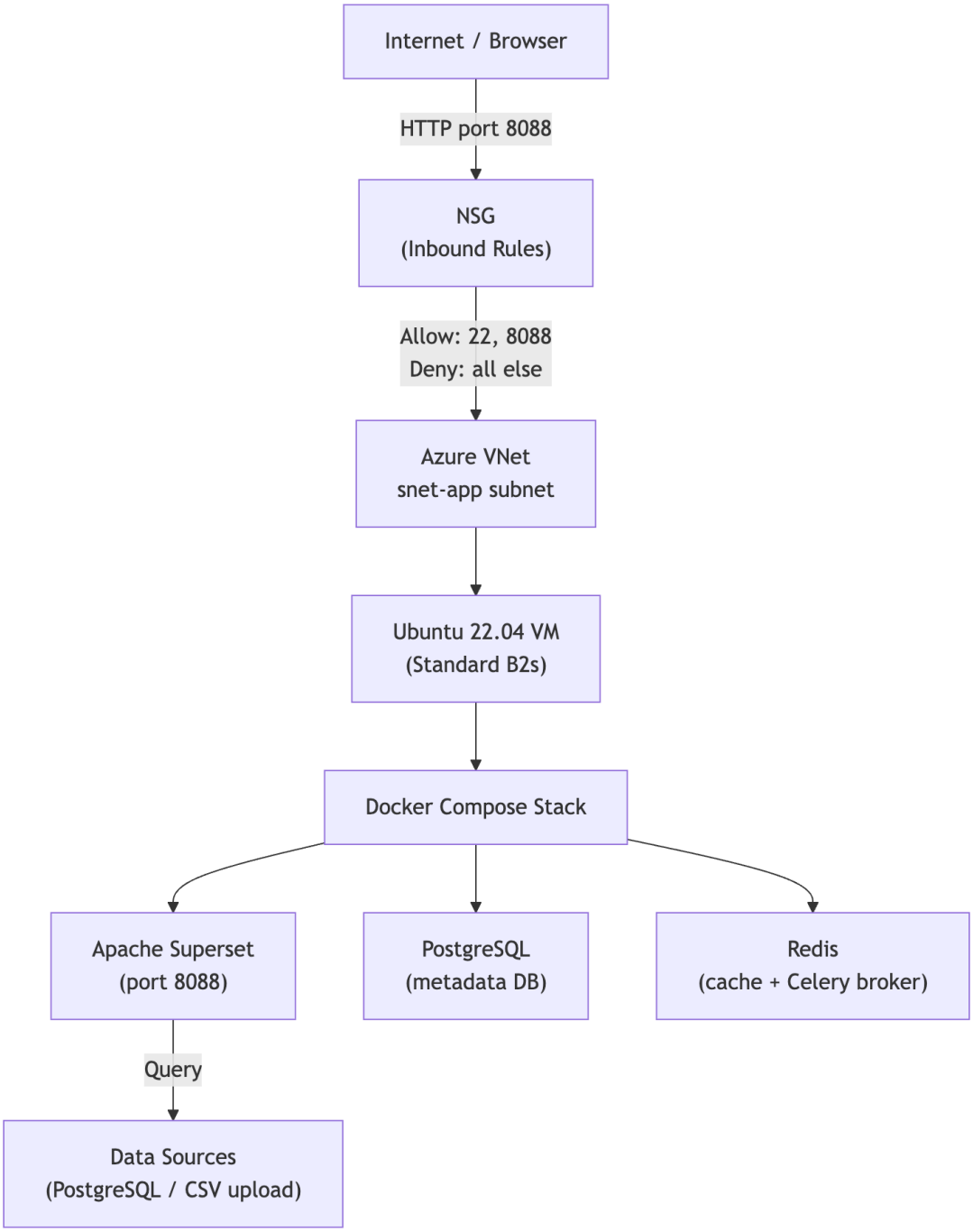


Figure 1: Azure deployment architecture – resource group, VNet, NSG, VM, and Superset Docker Compose stack.

2.3. Deployment Procedure

The deployment followed the account setup and four tasks specified in the assessment brief.

Task 0 – Account registration and Azure portal setup

An Azure account was activated at portal.azure.com using the student enrolment email. Azure's free account provides USD $200 in credits for 30 days alongside always-free services, covering the compute and networking requirements for this project (Microsoft, n.d.-d). The portal confirmed an active subscription in Australia East, providing the foundation for all subsequent resource provisioning.

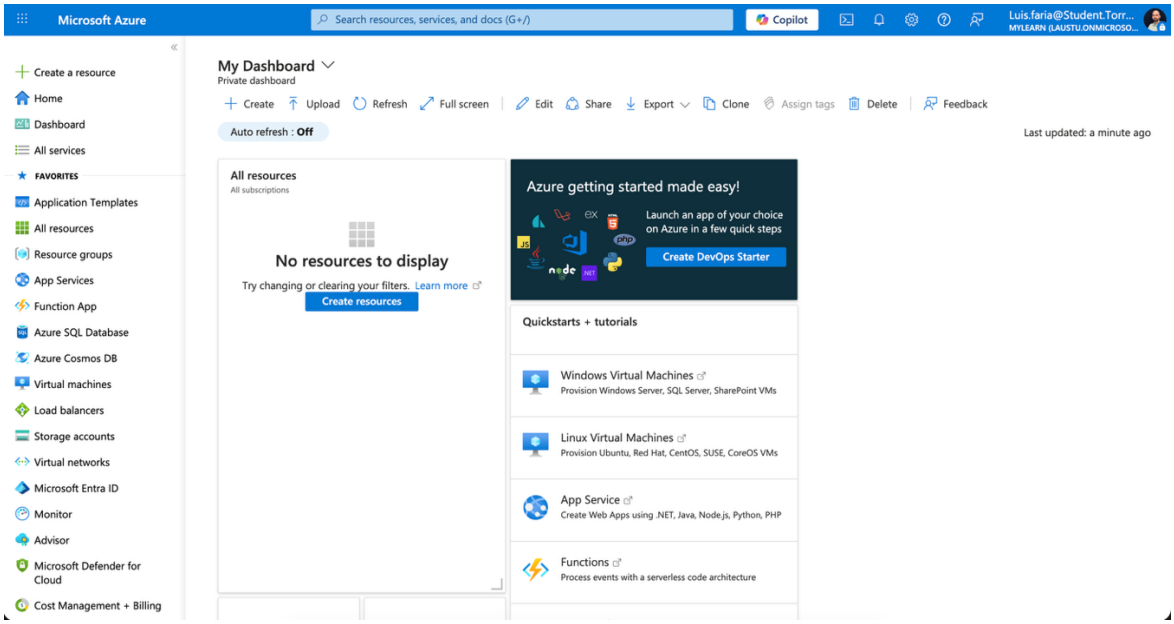
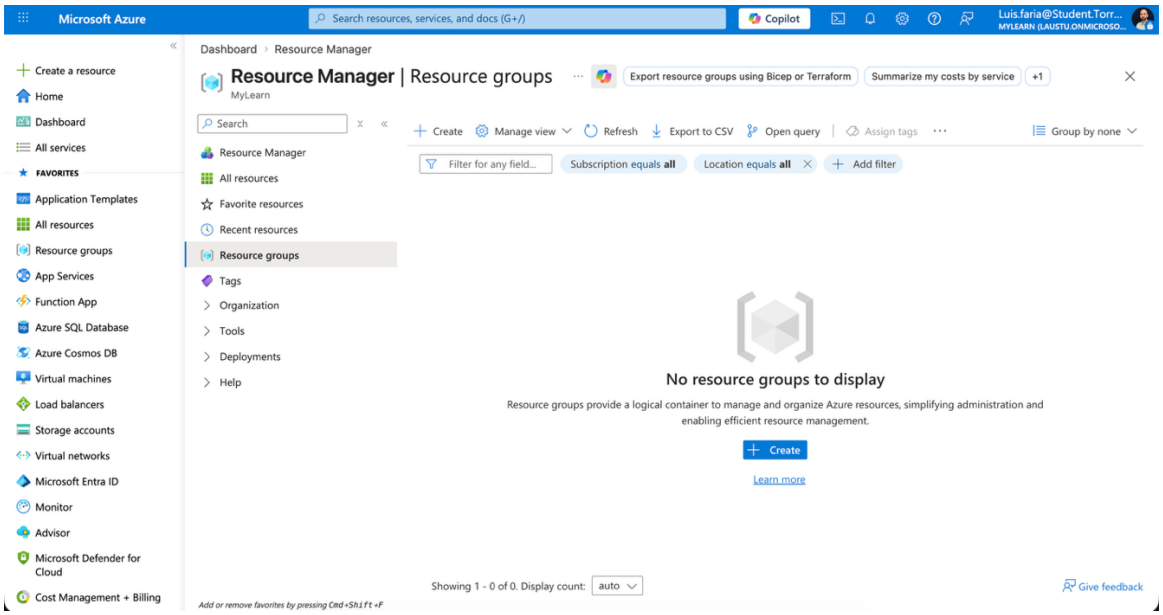
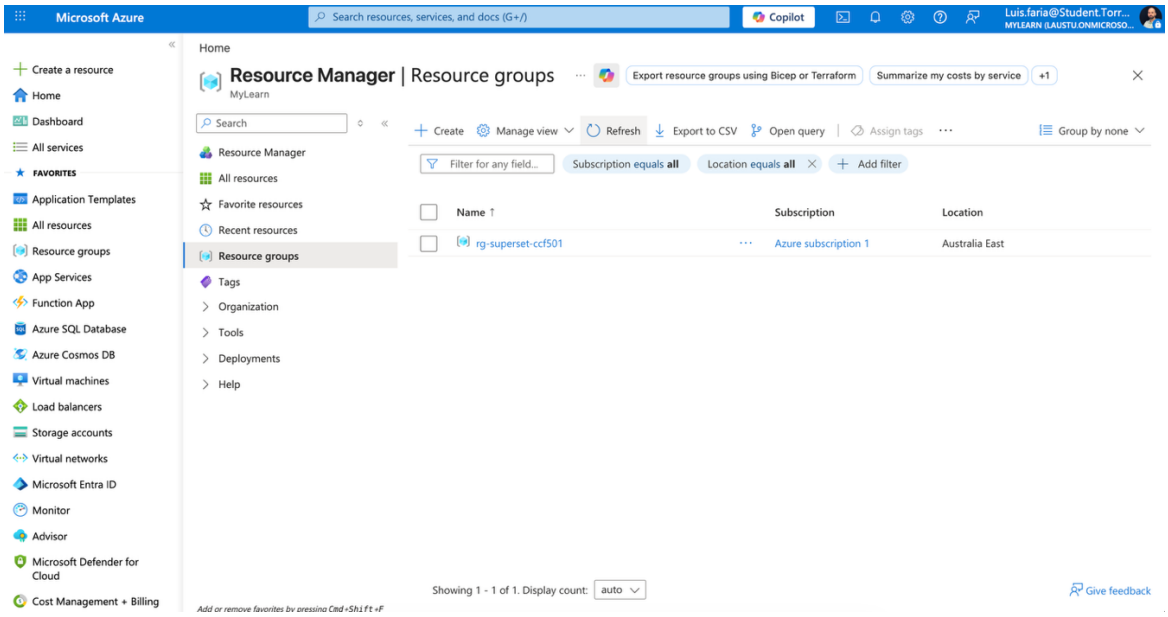


Figure 2: Azure portal - active subscription confirmed in Australia East.

Task a – Create a resource group

A resource group (*rg-superset-ccf501*) was created in the Azure portal under the Australia East region. Resource groups are logical containers for cloud resources, enabling unified billing, access control, and lifecycle management (Microsoft, n.d.-a). Australia East was selected to minimize latency and maintain local data residency.

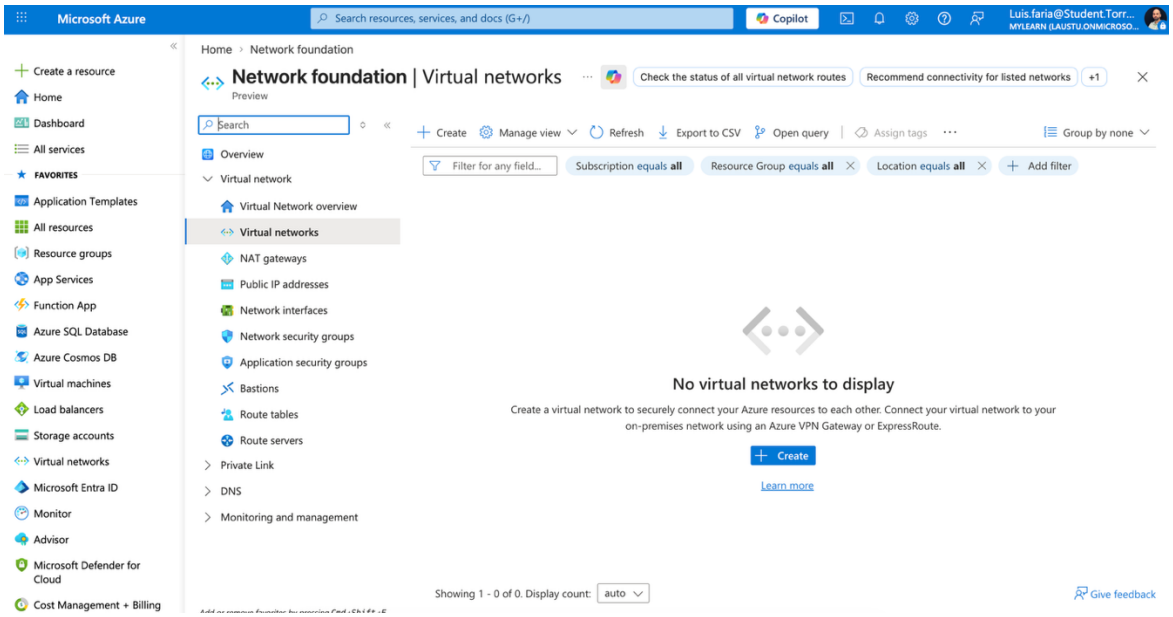




Figures 3A/3B: Before-and-after Azure resource group rg-superset-ccf501 in Australia East.

Task b – Add a virtual network

A Virtual Network (*vnet-superset*, address space *10.0.0.0/16*) was created with a dedicated application subnet (*snet-app*, *10.0.1.0/24*). VNets provides network isolation – resources internally without traversing the public internet (Microsoft, n.d.-b). This reflects NIST's resource pooling characteristic: shared physical infrastructure partitioned into isolated, private network boundaries (Mell & Grance, 2011).



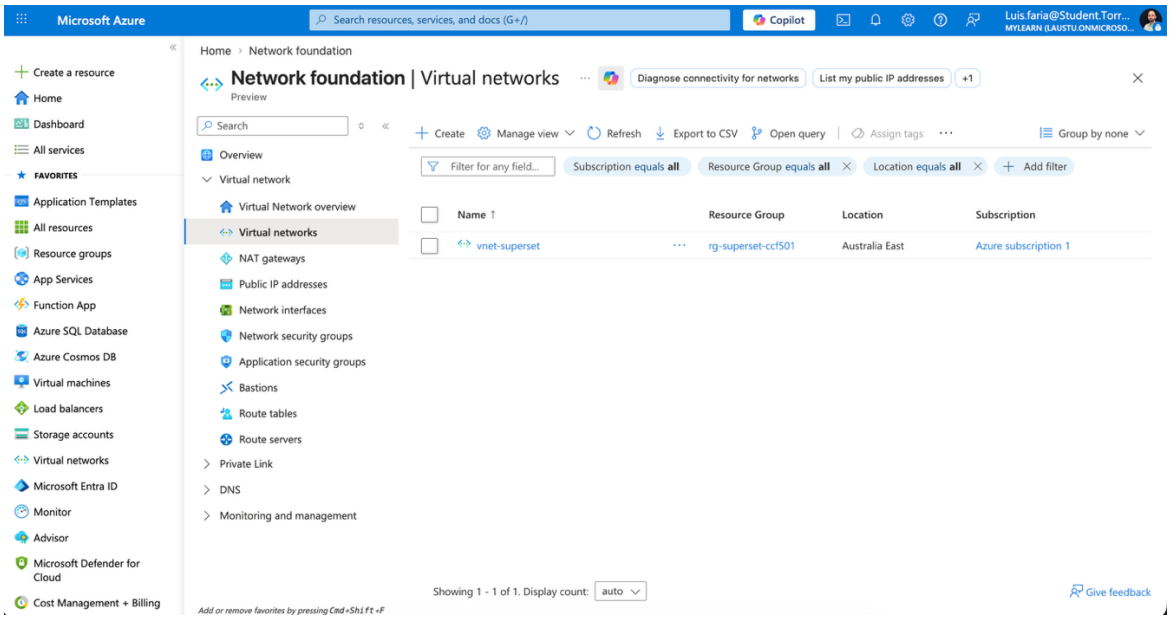
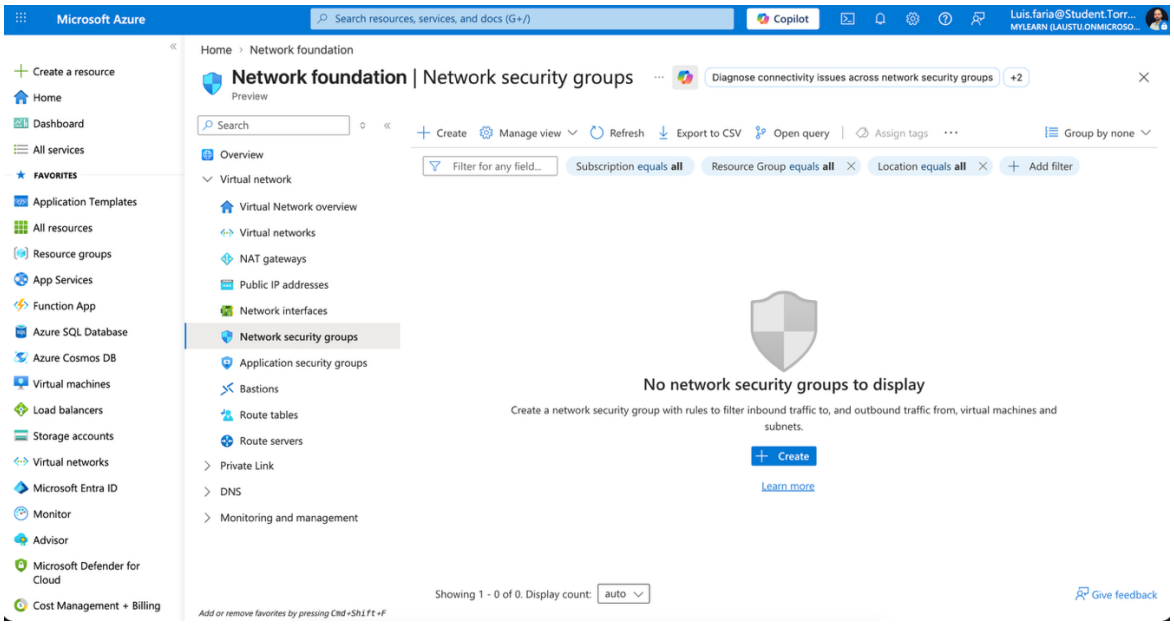


Figure 4A/4B: Before-and-after VNet vnet-superset with subnet snet-app.

Task c – Protect the network with a firewall / security policy

A Network Security Group (*nsg-superset*) was attached to the *snet-app* subnet. Inbound rules allow SSH (port 22, restricted to the author's IP) and the Superset port (8088); all other inbound traffic is denied by default. Ports 80 and 443 are intentionally excluded – this implements the principle of least privilege, exposing only the minimum required access (Shore, 2020).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Filter for any field...
Subscription equals all
Resource Group equals all
Location equals all
Add filter
No network security groups to display
Create a network security group with rules to filter inbound traffic to, and outbound traffic from, virtual machines and subnets.
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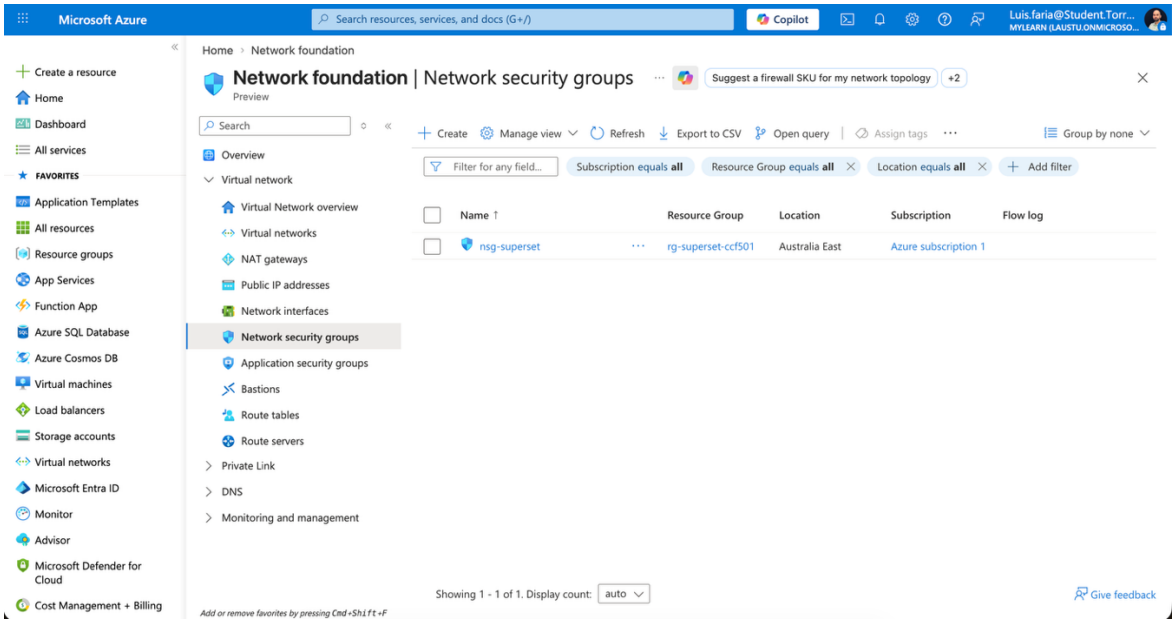
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Azure Cosmos DB
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Storage accounts
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Advisor
Microsoft Defender for Cloud
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Network security groups
Application security groups
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Private Link
DNS
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Add or remove favorites by pressing Cmd+Shift+F

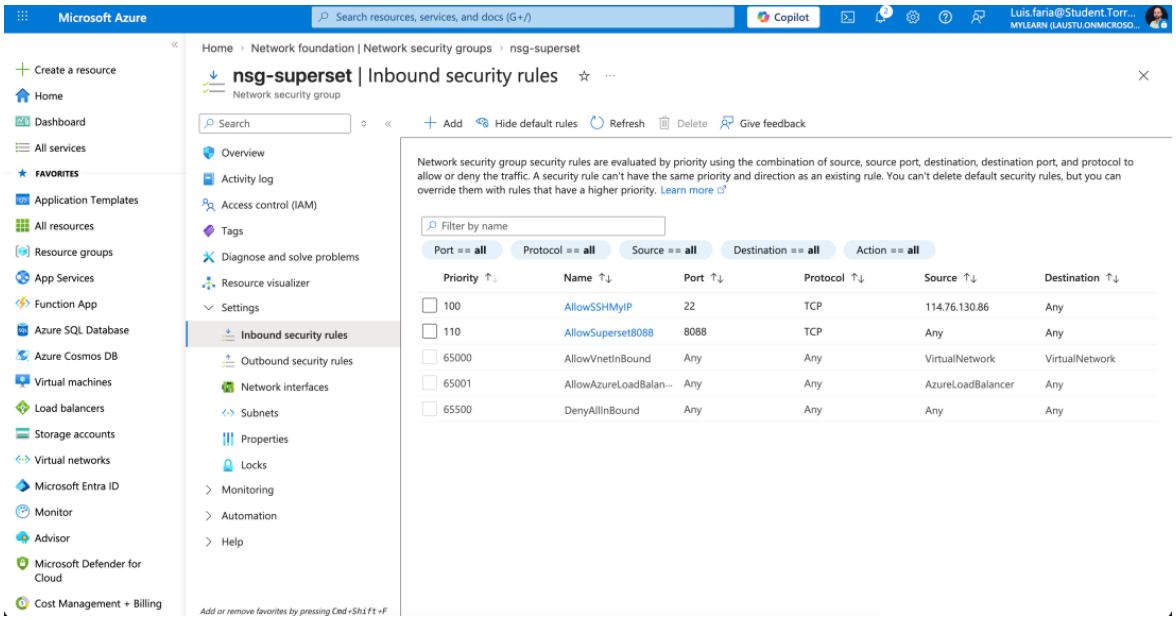

Figure 5A/5B/5C: Before-and-after NSG inbound rules – allow 22 (restricted to author IP) and 8088, deny all else.

Task d – Deploy Apache Superset

An Ubuntu 22.04 VM (Standard B2s: 2 vCPU, 4 GB RAM) was provisioned within *snet-app* and assigned a public IP. Docker and Docker Compose were installed via SSH, and the Superset Docker Compose stack was launched with *docker-compose up -d*. Superset was then accessible at *http://[PUBLIC_IP]:8088*; an admin account was created and a PostgreSQL data source connected to confirm end-to-end functionality.

Figure 6: Azure virtual machine overview showing Ubuntu host, running status, and public IP used for deployment access

Figure 7: Apache Superset login screen accessible at the Azure VM public IP on port 8088.

Figure 8: Apache Superset running successfully with a connected dataset and working analytics interface.

2.4. Cloud Services Security Policies

Security was addressed at three layers. At the **network layer**, the NSG enforces least-privilege access: port 22 is restricted to a single authorized IP, with all other inbound traffic blocked by default. At the **application layer**, Superset's RBAC provides three roles: Admin (full access), Alpha (create dashboards and run queries), and Gamma (view-only). At the **credential layer**, the secret key and database connection string are injected via environment variables – never hardcoded in source files.

Security Layer	Control	Threat Mitigated
Network (NSG)	Port allowlist + IP restriction on SSH	Unauthorized access, port scanning
Application (RBAC)	Admin / Alpha / Gamma roles	Privilege escalation, data exfiltration
Credentials	Environment variables (*.env* file)	Credential exposure in source control
OS	SSH key-based auth only (password disabled)	Brute-force attacks

Table 2: Security controls applied across network, application, credential, and OS layers.

Figure 9: Apache Superset RBAC role configuration.

2.5. Application Analysis and Robustness

The current deployment is functional but represents a minimal viable configuration. The primary limitation is a single point of failure: one VM hosts the entire stack – if the VM

becomes unavailable, the service is offline. A production-grade architecture would introduce the following improvements.

HTTPS / TLS termination: An Azure Application Gateway or Let's Encrypt certificate via nginx would encrypt traffic in transit. Currently, credentials are sent over HTTP.

Managed PostgreSQL: Replacing the Docker-hosted PostgreSQL instance with Azure Database for PostgreSQL provides automated backups, point-in-time restore, and high availability – eliminating the metadata loss risk if the VM disk fails.

Celery workers for async queries: Production Superset separates the web server from Celery workers for long-running queries, reusing the Redis broker already present in the Docker Compose stack. This pattern was previously applied by the author in Konquista – a WhatsApp automation platform built in Django + Celery + Redis that handled 1,000+ daily messages – where separating async workers from the web process eliminated UI blocking under load.

Azure Monitor + alerts: Attaching Azure Monitor to the VM provides CPU, memory, and disk metrics – enabling proactive alerting before resource exhaustion causes downtime.

Improvement	Azure Service	Benefit
HTTPS	Application Gateway / Let’s Encrypt	Encrypts traffic in transit
Managed DB	Azure Database for PostgreSQL	Automated backups, HA
Async queries	Celery + Redis (existing in Docker Compose)	Prevents UI blocking on long queries
Monitoring	Azure Monitor + Alerts	Proactive resource and availability alerting

Auto-scaling	Azure VM Scale Sets	Handles traffic spikes without manual intervention

Table 3: Proposed production improvements for the Superset deployment.

3. Conclusion

This report documented the deployment of Apache Superset, an open-source data exploration and visualization platform, on Microsoft Azure. The deployment demonstrated the four core tasks required by the assessment brief: provisioning a resource group and virtual network, enforcing a network security policy via a Network Security Group, and deploying a containerized open-source application on a cloud virtual machine. Each task was supported by screenshots and aligned with NIST's essential cloud characteristics – on-demand self-service in the provisioning process, resource pooling in Azure's shared infrastructure model, and measured service through consumption-based billing (Mell & Grance, 2011).

Microsoft Azure was selected for its data and analytics service ecosystem and its alignment with the author's AZ-900 and DP-900 certification pathway. The deployment reinforces the distinction between cloud and traditional IT infrastructure: resources that would require physical procurement and installation in an on-premises environment were provisioned, configured, secured, and running within [X] hours, at effectively zero capital cost. Areas identified for further investigation include TLS certificate automation, managed database integration for metadata persistence, and the separation of Celery workers for production-scale async query execution – improvements that would bring this deployment in line with enterprise-grade analytics infrastructure.

4. Appendices

4.1. Appendix A – Deployment Task Checklist

Task	Description	Status	Screenshot
0	Register Azure account / verify active subscription	AI / Software Startup	Figure 2
a	Create resource group (*rg-superset-ccf501*)	Google Cloud Platform (GCP)	Figures 3A-3B
b	Add virtual network (*vnet-superset* + *snet-app*)	IaaS + PaaS	Figure 4A-4B
c	Apply NSG inbound rules (22/80/8088)	Public cloud – multi-region	Figure 5A-5C
d	Deploy Apache Superset via Docker Compose	30-70% higher latency; engineers manually provisioning serves days before events	Figures 6-8

Table A1: Deployment task checklist with status and corresponding screenshots.

4.2. Appendix B – Glossary

Term	Meaning
IaaS	Infrastructure as a Service
NSG	Network Security Group
VNet	Virtual Network
RBAC	Role-Based Access Control
Docker Compose	Tool for defining and running multi-container Docker applications
Celery	Distributed task queue for Python applications

TLS	Transport Layer Security – encrypts data in transit
NIST	National Institute of Standards and Technology
AZ-900	Microsoft Azure Fundamentals certification
DP-900	Microsoft Azure Data Fundamentals certification
SPOF	Single Point of Failure

Table B1: Glossary of technical terms used in the report.

End of Appendix Section

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